USER GUIDE

Income and Substitution Effects

Change a price and the optimal bundle moves for two reasons at once. Compensation separates them: first pivot the constraint without changing welfare, then hand the income back.

BEFORE YOU START

Opening it

There is nothing to install. It is a web page: it opens in Chrome, Safari, Edge or Firefox, on a computer, tablet or phone. You need a connection the first time; after that the browser keeps it and it works offline.

Top right there are two pairs of buttons. The first switches the language between **ES** and **EN**; the second goes from **Dark** to **Light**. For projecting in class, light mode reads better on a beamer; on screen, dark is easier on the eyes.

> **Every graph zooms.** Mouse wheel over the graph, or a two-finger pinch on the trackpad or touchscreen. The zoom applies to the graph under the pointer only, not to the whole page.

> Each slider has a number box beside it. For an exact value, type it into the box and press `Enter`; the slider is for exploring, the box is for precision.

THE THREE BUNDLES

Why there have to be three

When p_x rises the consumer buys less x for two distinct reasons happening at once: x has become dearer *relative to* y, and the consumer is also poorer in real terms. The decomposition invents an intermediate bundle to pull them apart.

BUNDLE	WHAT IT IS
B_0	The original choice, at the starting prices and income.
B_S	The choice at the **new prices** but with **compensated** income: the consumer has seen relative prices change, but has not got poorer. From B_0 to B_S is the **substitution effect**.
B_1	The final choice, at the new prices and the real income. From B_S to B_1 is the **income effect**.

> The total effect, B_0 to B_1, is the sum of the other two. It is the only observable one: B_S does not exist in the data, it is an analytical device.

Slutsky or Hicks

Compensation can be defined two ways, and the applet has both because the textbooks do not agree.

RULE	WHAT IS COMPENSATED
Slutsky — hold the bundle	Give exactly enough to buy **the original bundle** at the new prices. It is observable — you only need to know what was bought before — and it leaves the consumer slightly *better off*, because on top of repeating the bundle he can rearrange it.
Hicks — hold the utility	Give exactly enough to reach **the original utility** at the new prices. Not observable, since nobody sees an indifference curve, but it holds welfare exactly fixed, which is what the theory asks for.

> Switching rule changes where B_S lands and therefore how the total splits between the two components. What does **not** change is the total effect, nor the sign of the substitution effect: it always points against the price, under either rule and with any utility function.

WHAT IS ON SCREEN

The two panels

PANEL	WHAT IT SHOWS
Plane (x, y)	The three constraints, the indifference curves, the three bundles and — if you switch the arrows on — the effect vectors.
Demand panel	Price on the horizontal axis and quantity demanded on the vertical, with **ordinary** and **compensated** demand overlaid. You can show x, y, or both at once.

> Showing **both** goods teaches the most. The effect on y is usually the surprise: when p_x rises, demand for y moves even though its own price did not change.

Layers and arrows

LAYER	WHAT IT DRAWS
Original / compensated / final constraint	The three lines. The compensated one is parallel to the final one and passes through B_0 (Slutsky) or is tangent to the original curve (Hicks).
Bundles (B_0, B_1)	Just the start and the finish, labelled. This is the honest view: what is observed.
Decomposition (B_S)	Adds the intermediate bundle. Switch it on when you are about to explain the split, not before.
Effect arrows	The vectors. Without the decomposition you get only the total effect, B_0 to B_1. With it, two more appear: a **magenta** one from B_0 to B_S labelled “Subst. effect” and a **mint green** one from B_S to B_1 labelled “Income effect”. The same colours mark the stretches on the x axis and on the y axis.
Indifference curves	The ones through the bundles.
Contours	The rest of the map, at regular levels.
Grid	The grid.

The readouts

READOUT	HOW TO READ IT
B_0 · original, B_S · substitution, B_1 · final	The three bundles, with their coordinates.
Compensated income	How much to hand over or take back for the chosen compensation.
Substitution / Income / Total	The three effects, in units of the good, for x and for y.

THE TEXTBOOK CASE, IN NUMBERS

Cobb–Douglas, price doubling

1. Pick **Cobb–Douglas** with a = **0.5**. Starting point: p_x = **1**, p_y = **1**, m = **8**. Change: Δp_x = **1**, Δp_y = **0**.

   ```
   B₀ = (4, 4) with u = 4. B₁ = (2, 4). The total effect on x is −2.
   ```

2. Look at y first: it has not moved. With Cobb–Douglas the spending on each good is a fixed share of income, so the rise in p_x does not touch y at all.

3. Set compensation to **Slutsky** and switch on **Decomposition (B_S)**.

   ```
   Compensated income 12 = 2·4 + 1·4: exactly what the old bundle costs at the new prices. B_S = (3, 6). Substitution −1, income −1.
   ```

4. Switch to **Hicks** without touching anything else.

   ```
   Compensated income 11.314 = 8·√2: just enough to get back to u = 4. B_S = (2.828, 5.657). Substitution −1.172, income −0.828.
   ```

5. Compare the two readings: the total is still −2 under both rules, but the split changes. Slutsky over-compensates — it leaves the consumer at u = 4.243, above the original — and so attributes less to the substitution effect.

6. Switch on **Effect arrows** and look at the axes: the **magenta** and **mint green** stretches on the x axis add up to the total move, and the same two stretches appear on the y axis, because y moves between B_0 and B_S even though it ends where it started.

What the applet warns about

If over some stretch of the sweep the **ordinary** demand for a good rises with its **own** price, a red warning appears naming the prices between which it happens. Only the own price counts: a cross-price curve sloping up merely means the goods are gross substitutes, which is ordinary and needs no warning.

A checked recipe

1. Pick the **X inferior** family. Parameters: $\bar{y}$ = **4**, β = **0.4**, $\underline{x}$ = **0.6**.

2. Starting point: p_x = **3**, p_y = **1**, m = **5.6**. Change: Δp_x = **2**, Δp_y = **0**. Leave the demand panel on **x**.

   The warning appears: “ordinary demand for x rises with its own price between 2.72 and 6.66”.

3. Look at the three bundles under **Slutsky**.

   B_0 = (0.644, 3.667), B_S = (0.615, 3.815), B_1 = (0.787, 1.667). Substitution −0.030, income +0.172, total **+0.142**.

4. There is the whole mechanism, in three numbers. Substitution is negative, as always. The income effect is **positive**, because x is inferior. And it is big enough to dominate: the total comes out positive and demand rises with the price.

5. Switch to **Hicks**: substitution −0.026, income +0.168, total +0.142 again.

   The split shifts a little, the total does not. That is the check that the decomposition is a device, not a fact.

6. Now look at the demand panel: the **ordinary** curve rises from left to right, and the **compensated** one falls. Both at once, in the same picture.

   Compensated demand can never slope upward. That is the law of demand that does always hold.

For the in-between case — x inferior but not Giffen — lower income to 3, below $p_y \cdot \bar{y}$ = 4. The warning disappears and the income effect is still positive, but no longer wins.

THE SLIDER LIMITS

What they are for

The **Slider limits** group has a pair of boxes per variable. They decide two things: how far each slider travels, and what stretch the demand panel sweeps. The defaults suit the textbook case, but hunting a Giffen nearly always needs them widened.

> The limits do not change the economics, only the window you look through. If an effect seems to vanish when you move a limit, it has left the frame, not stopped existing.

WRITING YOUR OWN FUNCTION

The syntax it accepts

In the custom preferences box the variables are **x** and **y**. With a hand-written utility, the optimal bundle and the Hicks compensated income are computed numerically, so functions with kinks are fine.

CONTROL	WHAT IT DOES
`+ − * / ^`	Add, subtract, multiply, divide, power. `x^0.5` is the square root of x.
`( )`	Brackets. `(x+y)^2` is not the same as `x+y^2`.
`sqrt ln log exp abs`	Root, natural log, base-10 log, exponential and absolute value.
`min max`	Two arguments: `min(x,y)` is perfect complements.
`sin cos`	Trigonometric, in radians. Rarely useful here, but available.
`e pi`	The two constants. `e` is 2.718…, `pi` is 3.1416…
`2x`, `3xy`	Implicit multiplication works: `2x` reads as `2*x`.

Examples that work

EXPRESSION	WHAT PREFERENCES IT DESCRIBES
`x^0.5*y^0.5`	The textbook case, the one that checks the numbers above.
`min(x,y)`	Perfect complements: the substitution effect is **zero**, and the whole move is income effect.
`x+y`	Perfect substitutes: the substitution effect takes everything, all at once.
`2ln(x)+y`	Quasilinear: the income effect on x is **zero**, so substitution and total coincide.

> If it cannot be read you get *“I can't read that expression”* and the graph stays as it was. The usual causes are an unclosed bracket, a decimal comma instead of a point (`0,5` instead of `0.5`) or a variable other than x and y.

IF SOMETHING GOES WRONG

Common problems

SYMPTOM	WHAT TO DO
The three bundles coincide	You have not changed any price. Move Δp_x or Δp_y.
B_S does not appear	Switch on the **Decomposition (B_S)** layer. The **Bundles** layer shows only B_0 and B_1 on purpose.
The substitution effect comes out positive	It cannot be, in the own price. If you see one, you are looking at the effect on the *other* good, which can go either way.
It says “Something runs outside the frame”	Raise the x or y ceiling in the view group, or tick **Fit the axes automatically**.
The demand panel comes out in pieces	Utility is undefined over part of the sweep — typical of the X inferior family, which needs $x > \underline{x}$ and $y < \bar{y}$. Narrow the p_x limits or change income.
The red flag never appears	The Giffen case is rare and has to be hunted. Use the recipe above, and remember it needs $m > p_y \cdot \bar{y}$.

Open the applet: Income and Substitution Effects. It all runs in the browser, and once loaded, offline too.

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